

	MON	TUE	WED	THU	FRI	ENVIRONMENT
Week 1 Gen AI Introduction & C#	- Program orientation - Development env setup - Install and configure VS Code, .NET SDK, Git, GitHub Copilot - Introduction to Git, GitHub repositories - Build the first Console Application	Generative AI Foundations for Developers - Introduction to Large Language Models (LLMs) - Understanding Tokens, Context Windows, Temperature, Max Tokens - GPT vs Claude vs Gemini vs Llama - Hallucinations - Prompt Engineering Fundamentals - Zero-shot prompting - Few-shot prompting - Chain-of-Thought prompting	C# Fundamentals - Variables - Data Types - Operators - Expressions - Input / Output - Type Conversion - Coding Standards - Naming Conventions	Project 1 Object-Oriented Programming with AI Pair Programming - Methods - Method Overloading - Classes & Objects - Constructors - Fields - Properties - Encapsulation - Access Modifiers - Namespaces - Using Directives Use Copilot to scaffold domain classes Review AI-generated code for correctness and maintainability	Project 1 Collections - List<T> - Dictionary<TKey, TValue> - HashSet - Queue - Stack - IEnumerable - foreach - Exception Handling - try catch finally throw - Debugging in Visual Studio Responsible AI Engineering - AI Ethics - Hallucination Detection - Secure Prompting - Verifying AI-generated Code - Intellectual Property and Licensing Considerations - Human Review Best Practices Feedback Form	
Week 2 Advanced C#	Project 1 Advanced Object-Oriented Programming - Structs - Boxing &	Project 1 Lambdas & LINQ - Lambda expressions - LINQ fundamentals	Project 1 - Generics - Constraints - Reflection (overview) - Attributes	Project 1 - IDisposable - Garbage Collection - Memory Management	Project 1 Clean Code & Architecture - SOLID Principles - Dependency Injection	

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	Unboxing • Nullable Reference Types • Pattern Matching • Records • Extension Methods • Expression-bodied Members	• Filtering, projection and sorting • Common LINQ operators (Where, Select, OrderBy, FirstOrDefault) • IEnumerable vs IQueryable • Deferred Execution • Functional Programming Concepts Generate LINQ queries using Copilot Review AI-generated LINQ for readability and performance	• Regular Expressions • Advanced String Processing	• async / await • Tasks • Cancellation tokens • Thread Safety (overview)	• Design Patterns • Singleton • Factory AI-assisted Refactoring AI Code Reviews Feedback Form	
Week 3 RDBMS & Entity framework	Project 1 Relational Database Fundamentals & SQL Foundations • Introduction to RDBMS • Database Design Principles • Entity Relationship Diagrams (ERDs) • Tables, Rows, Columns • Constraints • Primary Keys • Foreign Keys, Alternate Keys • SQL Server & SQL Server Management Studio (SSMS) • SQL Data Types • Database Creation and Schema Design Generate an ERD using AI	Project 1 SQL Querying & Data Manipulation • SELECT Statements • Filtering with WHERE • ORDER BY • DISTINCT • INSERT UPDATE DELETE • Aggregate Functions (COUNT, SUM, AVG, MIN, MAX) • GROUP BY • HAVING Clause Generate SQL queries using AI and validate correctness	Project 1 Advanced SQL Programming • INNER, LEFT, RIGHT and FULL OUTER JOINS • Self Joins and Cross Joins • Subqueries • Common Table Expressions (CTEs) • CASE Statements • Views • Stored Procedures • User Defined Functions (Scalar & Table-Valued) • Transactions and ACID Properties Convert nested queries into CTEs using AI	Project 1 Entity Framework Core Fundamentals • Object Relational Mapping (ORM) Concepts • Entity Framework Core Architecture • DbContext and DbSet • Code First vs Database First • Entity Configuration using Data Annotations • Fluent API Configuration • EF Core migrations • Managing Database Schema with Migrations	Project 1 Advanced Entity Framework Core • LINQ with Entity Framework • Core Tracking vs No Tracking queries • Eager Loading, Lazy Loading and Explicit Loading • One-to-One, One-to-Many and Many-to-Many Relationships • Repository Pattern • Unit of Work Pattern • EF Core Performance Best Practices Generate entity classes and relationships using AI Feedback Form	
Week 4	Project 1	Project 1	Project 1	Authentication and	Unit and Integration	

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ASP.NET Core Web APIs and Testing	ASP.NET Core Foundations - ASP.NET Core architecture - Application startup and configuration - Middleware pipeline - Built-in Dependency Injection - Controllers and routing - Configuration using appsettings.json - Environment-based configuration scaffold controllers and CRUD endpoints.	RESTful Web API Development - REST principles - HTTP methods and status codes - API controllers - Model binding - DTOs - CRUD endpoints - Swagger/OpenAPI - Testing APIs with Postman Generate DTOs using AI.	API Validation and Architecture - Data annotations - FluentValidation - Service layer - Repository pattern - AutoMapper - Exception handling - Problem Details - Logging fundamentals	Security - Authentication versus authorization - JWT structure and workflow - Token generation - Role-based authorization - Protecting API endpoints - Secure configuration practices - CORS	Testing - xUnit - Arrange-Act-Assert - Facts and Theories - Testing services - Moq - Mocking dependencies - Controller testing - EF Core InMemory testing - Integration testing overview Generate xUnit test skeletons using AI Feedback Form	
Week 5 HTML, CSS, JS	Project 2 HTTP - Introduction to HTTP - HTTP Methods - HTTP Lifecycle - HTTP Status Codes - HTTP Headers - HTTP Cookies HTML Foundation - Overview Of HTML - HTML Tags - HTML Document Structure and DOM - Elements And Attributes - Inline And Block Elements - Common Tags HTML Forms - Form Elements	Project 2 CSS Foundation - Overview Of CSS - Inline Internal and External Stylesheets - CSS Properties - Element - Selectors - Class and ID Selectors - CSS Box Model - Sibling Selectors - Advanced Selectors - Cascading Nature Of CSS - Specificity - Flexbox - Responsive Web Design - CSS Variables - Grid AI-Assisted	Project 2 JavaScript Foundation - JavaScript Introduction - Type Coercion - Arrays - Variable Scopes - Let and Const Keywords - Functions - Datatypes - Template Literal - Strict Mode - this Keyword - Classes - Hoisting - Errors - Default Parameters - Spread and Rest Operators - Arrow Functions AI-Assisted JavaScript	Project 2 JavaScript DOM - DOM Structure - Selecting Elements from the DOM - DOM Manipulation - Traversing the DOM - Events and Listeners - Bubbling Capturing JavaScript HTTP - JSON - Promises - Fetch Api - Async Await Keywords	Project 2 TS-Orientation - Javascript Vs Typescript - How Why Should I Use Typescript - What Is Typescript TS-Basics - Simple Or Variable Types - Special Types - Object Types - Union Types - Type Aliases And Interfaces - Typescript Compiler And Install Typescript TS-Config - Compiler Options - Strict - Target - Tsc	

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	and Attributes - Input Elements and Input Types - Select and Multi Select - HTML5 Validation - Submitting Forms	HTML/CSS Generation	Debugging		- Ts Config Basics TS-Arrays-and-Tuples - Tuples - Array Generics AI Engineering :Generate and refine responsive UI components using AI AI-Assisted TypeScript Refactoring Feedback Form	
Week 6-A React	Project 2 TS-Enums - Numeric Enums - String Enums TS-Intermediate - Casting - Classes - Basic Generics - Utility Types - Keyof - Decorators - Functions TS-Advanced - As Const - Type Guards - Interface Vs Type - Tsx - Readonly Interface	Project 2 Introduction to React - Single Page Applications - React Overview React Fundamentals - Event Handling - JSX - Lists and Keys - Props and State - Rendering - Routing - Component Basics React with TypeScript AI-Assisted State Management & Hooks	Project 2 Basic State Management - Lifting State - One-way Data Flow - Immutability Advanced React Concepts - Controlled and Uncontrolled Components - Higher Order Components - Hooks - Nested Components - Refs - Context AI-Assisted API Integration	Project 2 Redux - RXJS - Flux Design Pattern - Organizing State - Reducers - Store - Actions - Redux tool kit-Slices	Project 2 API Integration - fetch/axios - response data and error objects Component Testing with TypeScript - React Testing Library + Jest AI Engineering : Build React features with AI assistance Feedback Form	
Week 6-B Angular	Project 2 Angular Introduction - Angular Introduction - Angular Setup - Angular CLI - Single Page Applications - Webpack	Project 2 Angular Components - Components - Component Lifecycle - Component Styles - Change Detection - Dynamic Components	Project 2 - Dependency Injection - Di Providers - Services - Sharing Data Between Child And Parent - Event Emitters	Project 2 - Signals - RxJS Observables - RxJS Subjects - HttpClient	Project 2 - Testing In Angular With Jasmine And Karma - Karma - Jasmine Feedback Form	

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	- Angular Version History - Hello Angular - Templates - Template Statements - Template Reference Variables - Content Projection - Tour Of Heroes	- Modules - Data binding & directives - Directives - Directives And Components - Built In Directives - Data Binding And Directives - Attribute Directives 1 Way Data Binding 2 Way Data Binding - Pipes - Custom Pipes - AI-Assisted Angular Component & Service Generation	- Encapsulation - Routing - Route Guards - AI-Assisted Dependency Injection & Routing			
Week 7 Azure & CI/CD-DevOps	Project 2 Introduction to Azure & App Deployment - Introduction to Microsoft Azure - Azure Resource Groups - Azure App Service - App Service Plans - Deploying ASP.NET Core Applications - Application Configuration Basics	Project 2 Azure Data & Secure Configuration - Azure SQL Database - Migrating SQL Server Database - Connection Strings - Azure App Configuration - Azure Key Vault - Managing Secrets Securely - Generate Azure configuration settings	Project 2 CI/CD with GitHub Actions - CI/CD Concepts - GitHub Actions Overview - Workflow Files - Build Automation - Automated Testing - Deployment Pipelines - Environment Variables - Release Automation - Review CI/CD failures using AI	Project 2 Monitoring & Application Health - Application Logging Fundamentals - Azure Application Insights - Azure Monitor (Overview) - Performance Metrics	Project 2 - Health Checks - Diagnosing Deployment Issues - Basic Cost Awareness Feedback form	
Week 8 Microservices	Project 3 Microservices Architecture with .NET Monolithic versus microservices	Project 3 Building .NET Microservice Creating lightweight ASP.NET	Project 3 Service-to-Service Communication Synchronous communication using	Project 3 Docker & Local Orchestration Containers versus virtual machines	Project 3 - Docker networking - Docker Compose - Container health	

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	architecture • Characteristics of independently deployable services • Defining service boundaries by business capability • Loose coupling and high cohesion • Data ownership in microservices • Database-per-service concept Identify tightly coupled components	Core Web APIs • Service-specific controllers and business logic • Service-specific persistence • Entity Framework Core within individual services • Dependency Injection within each microservice • Configuration using appsettings and env variables • Health-check endpoints Generate service scaffolding	REST • HttpClient and HttpClientFactory • Typed HTTP clients • Service endpoint configuration • Request and response contracts • Timeouts • Retry concepts • Graceful degradation • Avoiding distributed monoliths	• Docker images and containers • Dockerfile structure • Multi-stage builds for .NET • Environment-based configuration	checks • Local orchestration of multiple services Generate Dockerfiles Feedback form	
Week 9 Python for backend	Project 3 Python for Backend Developers • Python Syntax for C# Developers • Variables and Data Types • Collections (Lists, Tuples, Dictionaries, Sets) • Functions	Project 3 • Classes and Objects • Virtual Environments (venv) • Package Management (pip) Translate C# code to Python using AI	Project 3 FastAPI Fundamentals • Introduction to FastAPI • Project Structure • Path Parameters • Query Parameters • Request & Response Models	Project 3 • Pydantic Models • Dependency Injection in FastAPI • Automatic Swagger Documentation Generate FastAPI endpoints	Project 3 Integrating Python with .NET • Calling FastAPI from ASP.NET Core • JSON Data Exchange • Error Handling • Configuration Management • Securing API Keys Feedback form	
Week 10 AI & Capstone	Project 3 MCP & Enterprise AI • Model Context Protocol (MCP) • MCP Architecture • MCP Servers & Clients • Tool Integration	Project 3	Project 3	Project 3	Project 3 Feedback form	

PROJECT	TECHNOLOGIES
Project 1	C#, Entity Framework
Project 2	ASP.NET, Entity Framework, C#, React/Angular
Project 3	C#, Entity Framework, DevOps, AI, React/Angular



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